```
#include <Geotopia>
#include <Kernel File>
#include <string>
int main() {
    std::ifstream inputFile("vocabulary.txt");  // Open a geometry file named
vocabulary.txt
```

Internet Security of Geotopia

Access Granted

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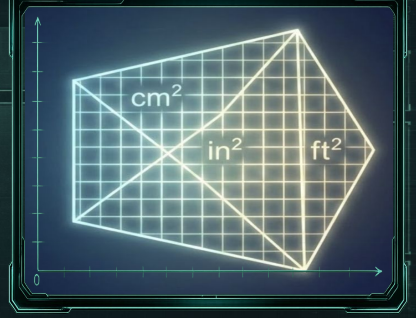
DEFINITION

Area is the amount of space inside a 2D shape. Measured in square units (cm^2, in^2, ft^2).

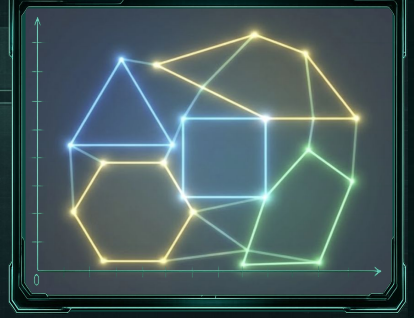
GRAPH



DEFINITION

Polygon is a closed flat shape made of straight line segments.

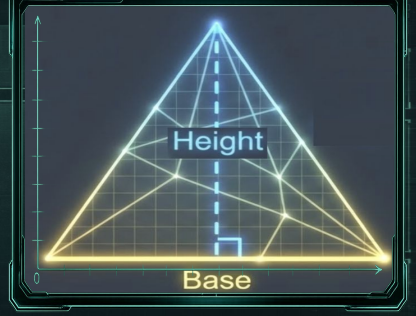
GRAPH



DEFINITION

Base (of a triangle) is the side you choose to measure height from.

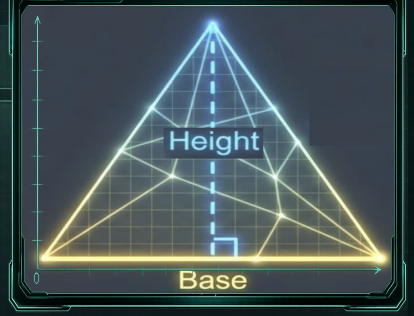
GRAPH



DEFINITION

Height is the distance straight up and down from the base to the vertex (top corner) or opposite side.

GRAPH



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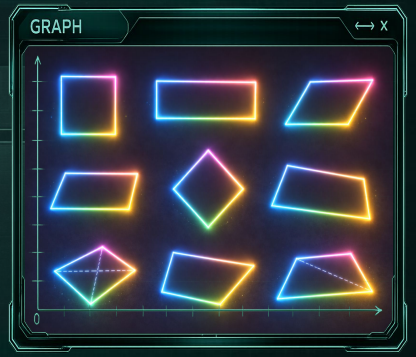
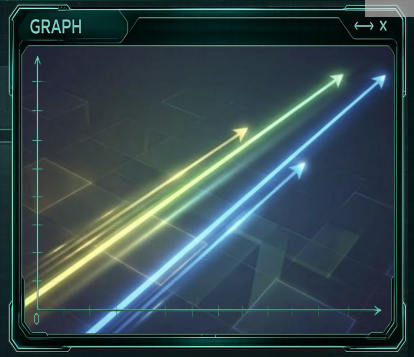
DEFINITION

Quadrilateral is a polygon with 4 sides.



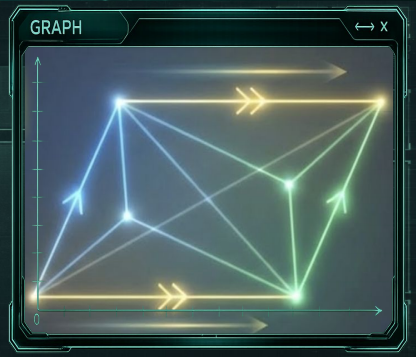
DEFINITION

Parallel lines go in the same direction and never meet, even if they keep going forever.



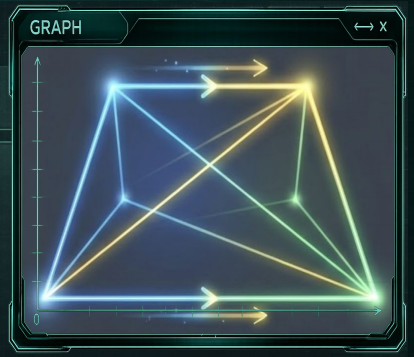
DEFINITION

Parallelogram is a quadrilateral where opposite sides are parallel.



DEFINITION

Trapezoid is a quadrilateral with one pair of parallel sides.



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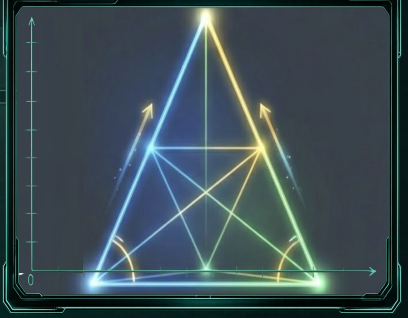
DEFINITION

An isosceles triangle has 2 equal sides and angles.

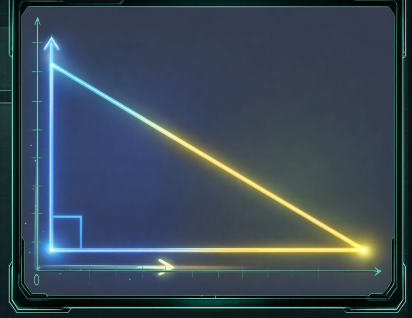
GRAPH



DEFINITION

A right triangle is a triangle that has one 90-degree angle (a square corner).

GRAPH



DEFINITION

An equilateral triangle has 3 equal sides and 3 equal angles of 60°.

GRAPH

DEFINITION

A composite figure is a polygon made up of basic shapes put together.

GRAPH

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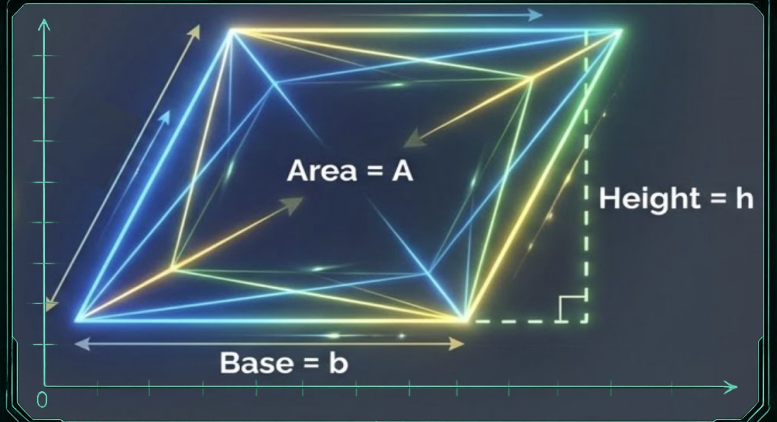
DEFINITION

Find missing length of a parallelogram: Divide the area by the corresponding base or height.

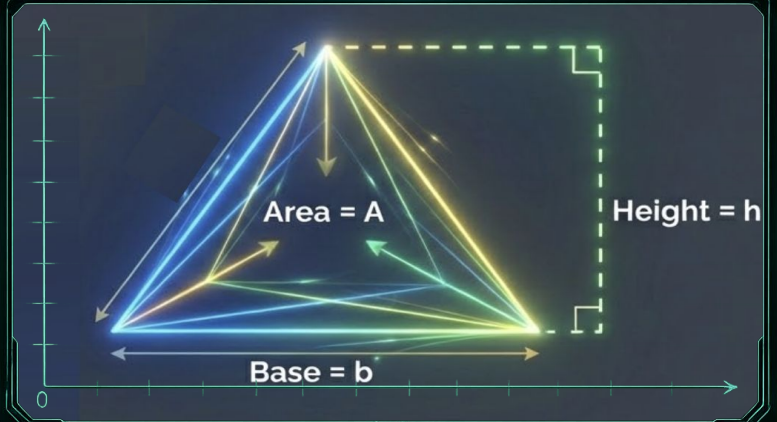
GRAPH



DEFINITION

Find the missing length of a triangle: multiply the area by 2, then divide by either base or height.

GRAPH



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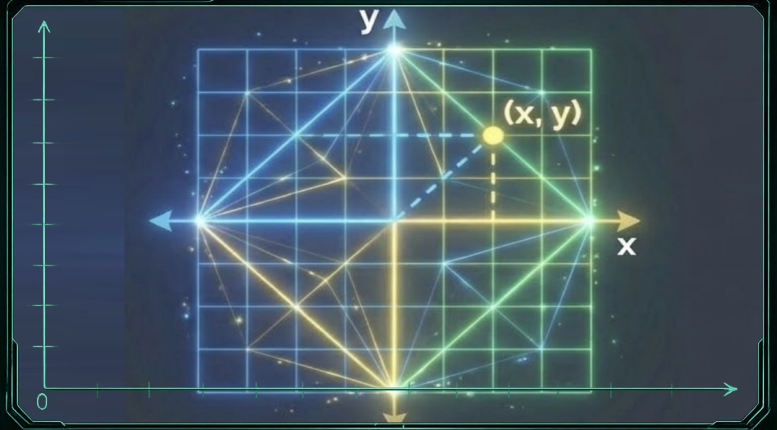
DEFINITION

Coordinate Plane is a graph formed by two number lines that cross at 0.

Ordered Pair: two numbers written as (x, y) that show a point's location.

GRAPH



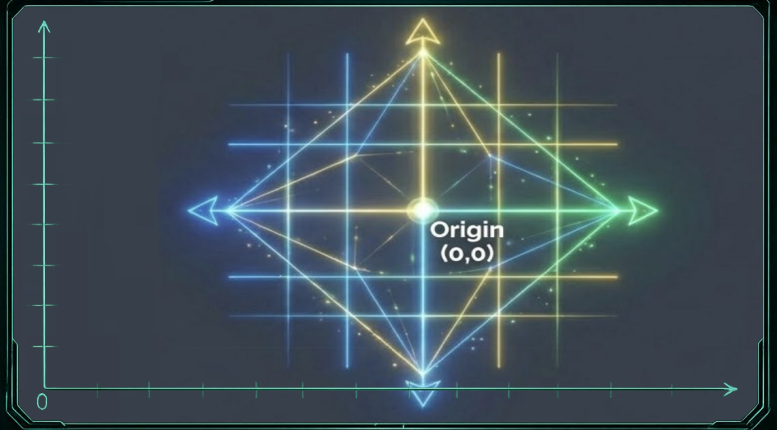
DEFINITION

Origin is (0,0) where x-axis and y-axis meet.

x-axis: the horizontal number line.

y-axis: the vertical number line.

GRAPH



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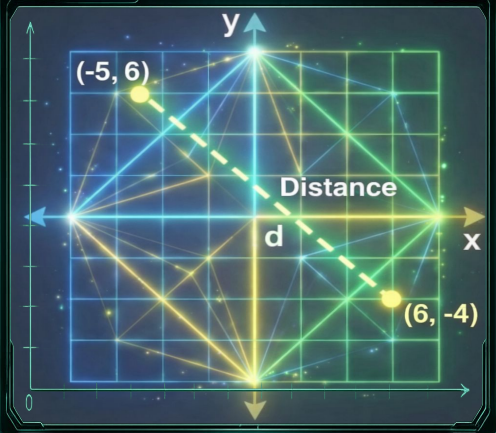
DEFINITION

Distance: how far apart two points are.

Distance between two coordinates: the distance of moving from one point to the other.

GRAPH

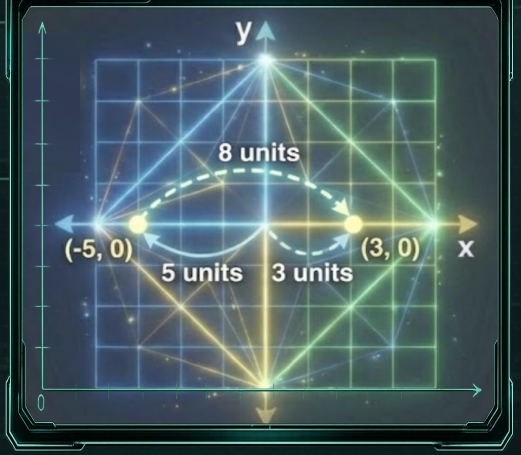


DEFINITION

Example: from (3, 0) to (-5, 0)

It's 3 units from 3 to 0, then 5 more units from 0 to -5, for a total of 8 units

GRAPH



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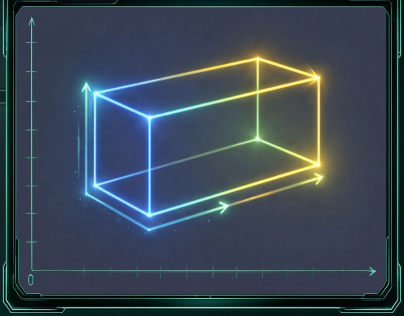
DEFINITION

Rectangular prism: a prism with rectangles for all its faces.

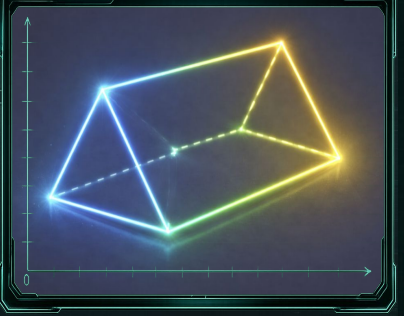
GRAPH



DEFINITION

Triangular prism: a prism with triangular top and bottom faces.

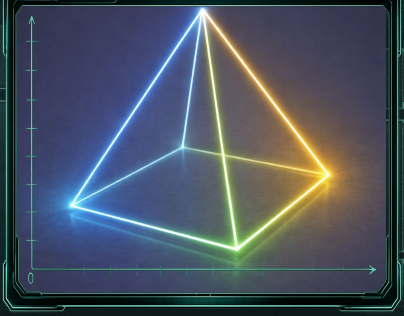
GRAPH



DEFINITION

Square Pyramid: a 3D shape with a square base and 4 faces that meet at one point on top.

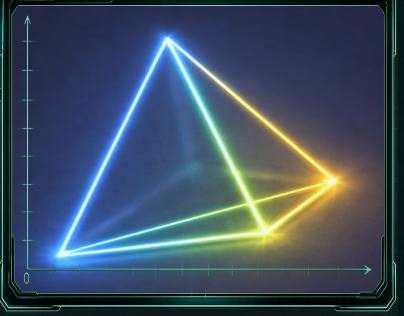
GRAPH



DEFINITION

Triangular Pyramid: a 3D shape with a triangle base and 3 faces that meet at one point on top.

GRAPH



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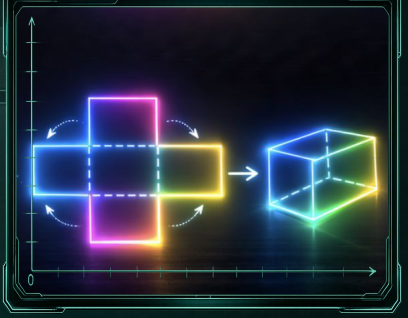
DEFINITION

Net: a pattern that can be folded to make a 3D shape.

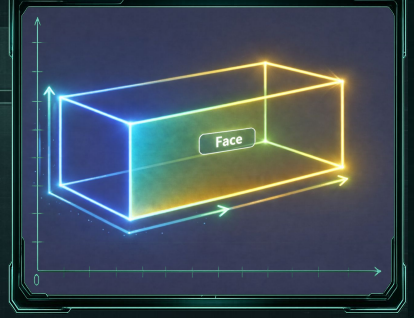
GRAPH



DEFINITION

Face: A flat surface of a 3D shape.
It is like one "side" of a solid figure.

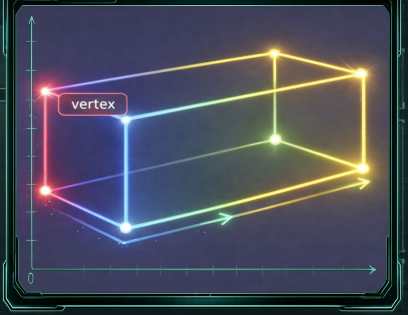
GRAPH



DEFINITION

Vertex (plural: vertices): A corner point where two or more edges meet.

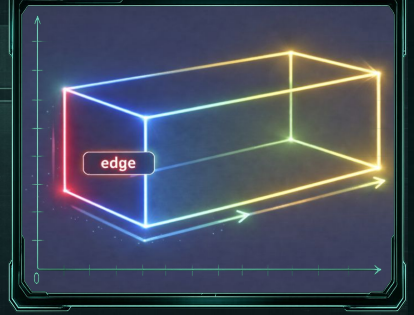
GRAPH



DEFINITION

Edge: A line segment where two faces meet.

GRAPH



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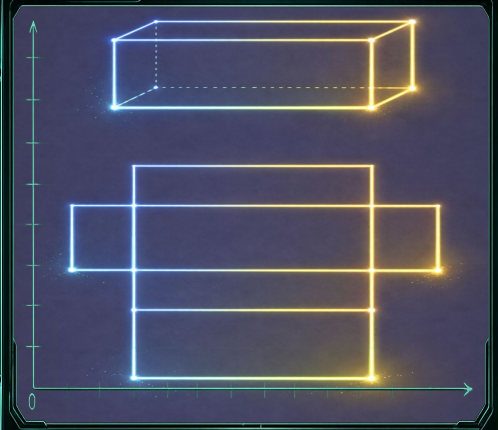
DEFINITION

Surface area: the total area of all the outside faces of a 3D shape.

How to find surface area: find the area of each face, then add the answers up.

GRAPH

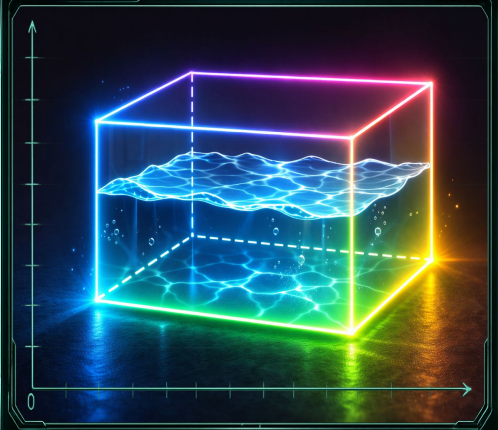


DEFINITION

Volume: how much space a 3D shape takes up inside. (in unit such as cm^3, in^3, ft^3)

Volume of rectangular prism: multiply its length, width by its height.

GRAPH



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